Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
4th Year 1st Semester Final Examination, 2022
Course Code: EEE-475, Session: 2018-19
Course Title: RF and Microwave Engineering

Full Marks: 100 **Credit: 3.0** **Exam Duration: 3 Hours**

Part-A

*[Answer any **two** of the following questions]*

1. (a) What is aperture of an antenna? Show how antenna aperture can be set to increase directivity of an antenna. 6
   (b) Calculate the half power beamwidth, beamwidth between first nulls, radiated power and directivity of an antenna with power pattern $\cos^2\theta \sin\varphi$ for $0 \leq \theta \leq \pi$, $0 \leq \varphi \leq \pi$. 6
   (c) Describe the field regions surrounding an antenna. 5
   (d) For an elemental magnetic dipole derive the equation for electric field and magnetic field intensity. 8

2. (a) Describe how passive remote sensing is used to detect object. 5
   (b) Explain how the RADAR range equation is used to detect objects. 6
   (c) Show the signal to noise ratio of a radio link is related to bandwidth and system temperature. 5
   (d) Derive the equation for a uniform linear array and write the conditions for main beam direction, width of main beam and null locations. 9

3. (a) Write short notes on.
      (i) Lorentz reciprocity theorem.
      (ii) Image theory. 6
   (b) Write the steps to determine line input impedance using Smith Chart. 4
   (c) A lossless transmission line length 0.434λ and characteristic impedance $100\ \Omega$ is terminated in an impedance $260 + j180\ \Omega$. Find the voltage reflection coefficient, standing wave ratio and the location of maximum voltage on the line. 7
   (d) A $50\ \Omega$ transmission line is connected to a load impedance $35 - j47.5\ \Omega$. Find the position and length of a short circuited stub required to match the line. 8

Part-B

*[Answer any **two** of the following questions]*

4. (a) Sketch the equivalent circuit of a high frequency resistor. Also plot a typical frequency versus impedance cure [*sic*: should be curve] of the resistor. 6
   (b) What is microstrip antenna? Mention some of the applications of microstrip line. 6
   (c) Derive the equation for a short circuit terminated transmission line and sketch voltage, current and impedance as a function of line length. 7
   (d) How is quarter wavelength transmission line used for impedance matching? 6

5. (a) For high pass filter determine ABCD parameters of a two port system. 6

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(b) What are stability circles? Explain how it is used. 5

(c) Explain the working principle of negative resistance oscillator and feedback oscillator. 7

(d) A lossless transmission line of length 1.5 m, which is less than quarter wavelength has open circuit impedance $-j54.6\ \Omega$ and short circuit impedance $j103\ \Omega$. Find Z_o and γ. Also find the input impedance if the line length is doubled and the length of line if it is to appear as an open circuit at the input. 7

6. (a) Write short notes on the following topics.

(i) Poincare Sphere
(ii) Yagi-Uda antenna. 6

(b) For a paraboloid-of-revolution with an isotropic point source what is the ratio of power density and field intensity ratio. 6

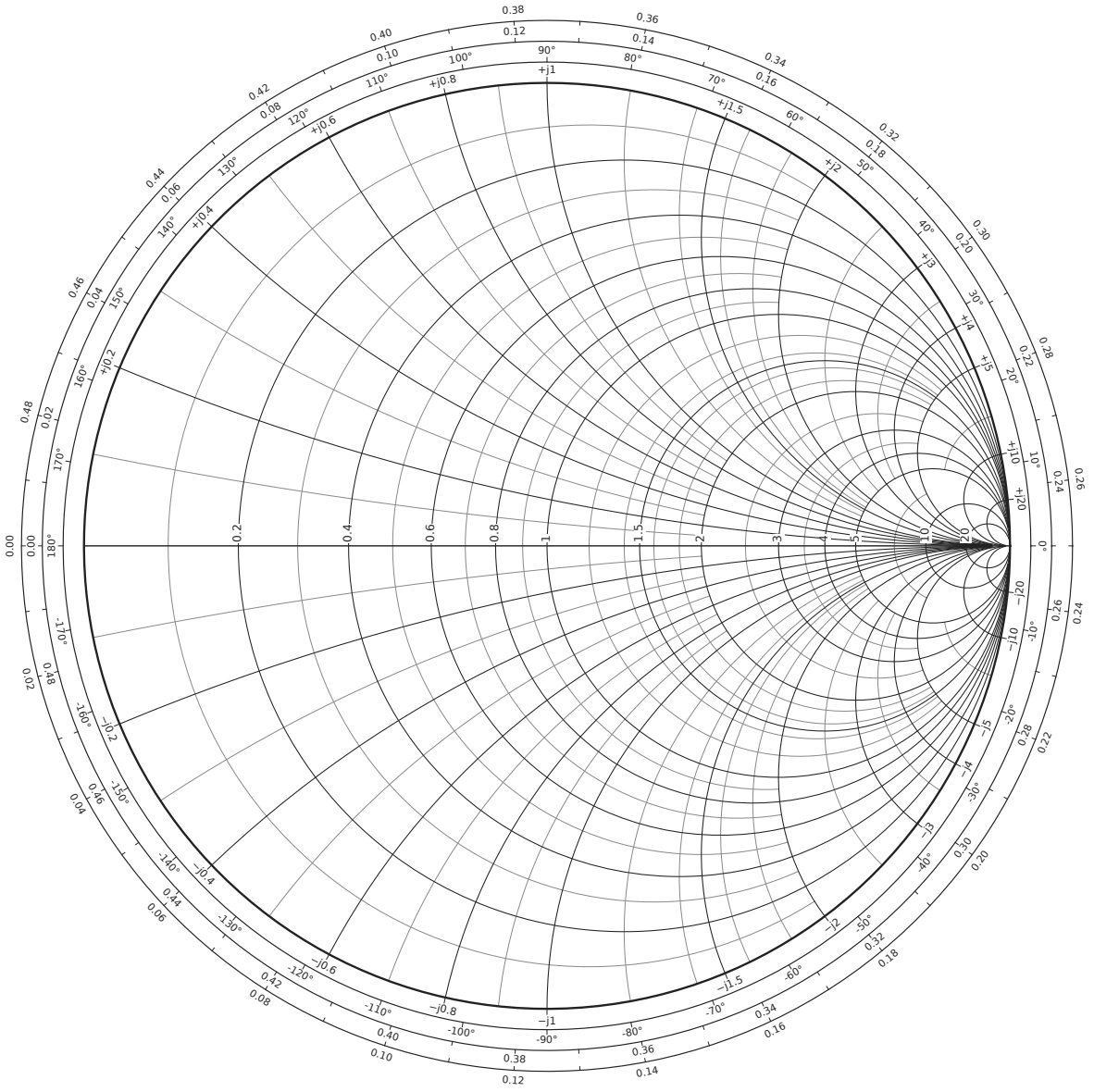
(c) Show that for antennas with same aperture the received to transmitted power ratio is inversely proportional to square of wavelength and distance between them. 6

(d) For a pyramidal or rectangular horn show that directivity D is $D \approx 10 \log\left(\frac{7.5 A_p}{\lambda^2}\right)$. 7

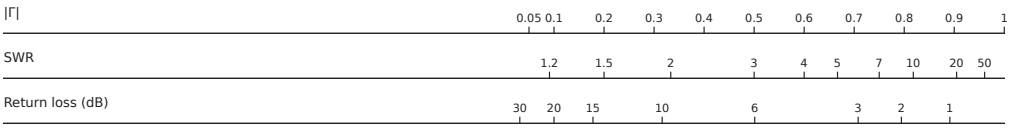
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Smith Chart

WAVELENGTHS TOWARD GENERATOR (inner) · TOWARD LOAD (outer)



ANGLE OF REFLECTION COEFFICIENT IN DEGREES (innermost ring)



Radial scales: read the distance from the chart's centre, with the centre at the 0 end of each scale.

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